Example: Sales of 5 different products

Imagine a shop sells laptops, mobiles, headphones, keyboards, and mice.

Product	Units Sold	Revenue (₹)
Laptop	10	5,00,000
Mobile	25	3,75,000
Headphones	40	80,000
Keyboard	30	45,000
Mouse	50	25,000

The 4 types of data analysis

1. Descriptive analysis — What happened?

Laptop sales generated the highest revenue: ₹5,00,000.

We summarize past data using totals, averages, charts, and percentages.

2. Diagnostic analysis — Why did it happen?

Why did laptops generate the most revenue? Because their selling price per unit is much higher than the other products in this example.

We investigate reasons, relationships, and causes.

3. Predictive analysis — What might happen next?

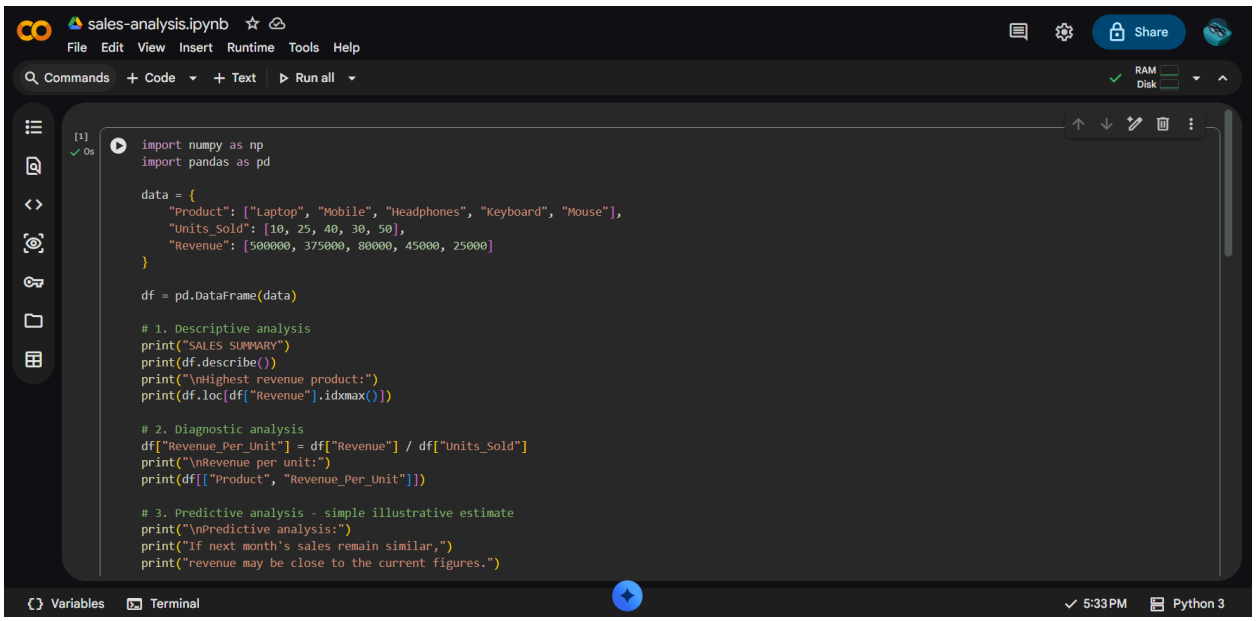
Based on previous sales trends, we might predict that laptop revenue will remain high next month.

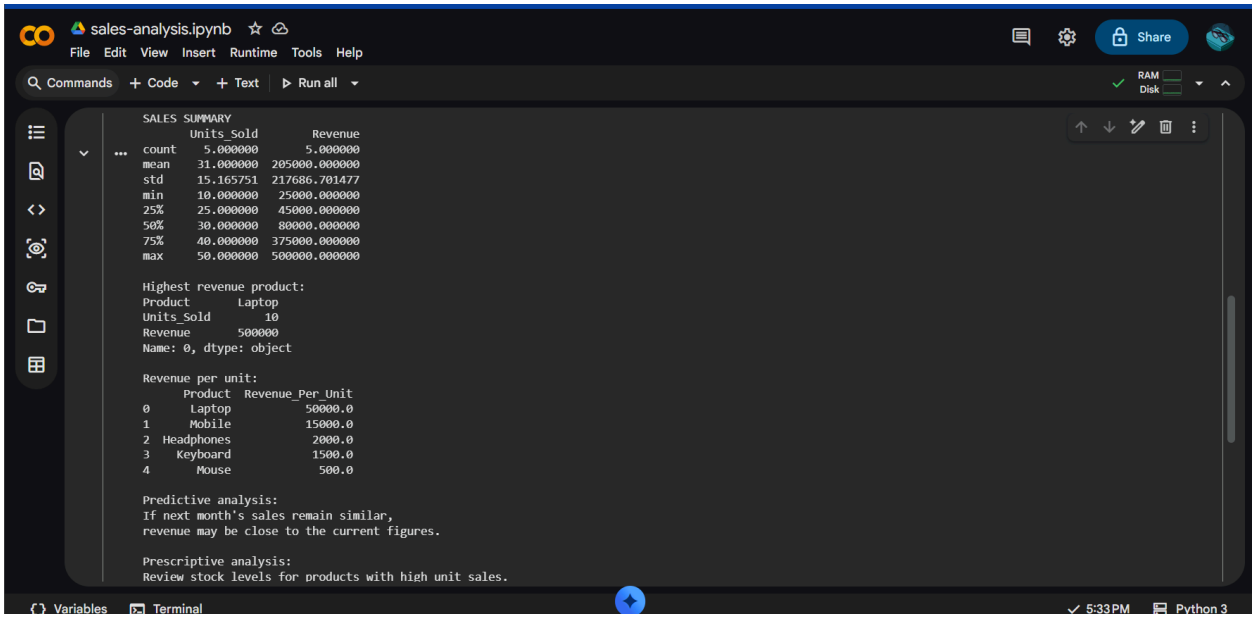
We use historical data and statistical or machine-learning models to forecast future outcomes.

4. Prescriptive analysis — What should we do?

Keep sufficient laptop stock and consider promotions for headphones if the goal is to increase their revenue.

We recommend actions based on predictions, goals, and constraints.




sales-analysis.ipynb
File Edit View Insert Runtime Tools Help
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RAM
Disk
SALES SUMMARY
Units_Sold Revenue
count 5.000000 5.000000
mean 31.000000 205000.000000
std 15.165751 217686.701477
min 10.000000 25000.000000
25% 25.000000 45000.000000
50% 30.000000 80000.000000
75% 40.000000 375000.000000
max 50.000000 500000.000000
Highest revenue product:
Product Laptop
Units_Sold 10
Revenue 500000
Name: 0, dtype: object
Revenue per unit:
Product Revenue_Per_Unit
0 Laptop 50000.0
1 Mobile 15000.0
2 Headphones 2000.0
3 Keyboard 1500.0
4 Mouse 500.0
Predictive analysis:
If next month's sales remain similar,
revenue may be close to the current figures.
Prescriptive analysis:
Review stock levels for products with high unit sales.
Variables
Terminal
5:33 PM
Python 3